



Comparative Analysis of Neurological Signals for Exoskeleton Control

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Abstract — Neurorehabilitation is a medical process that aims to recover the nervous system of patients with neurological conditions, costing US\$70,601. Major contributors to this high cost are the stay-related costs which total US\$26,340, effectively compromising 37.3% of the total rehabilitation cost. Lower limb exoskeletons (LLEs) have been developed to support rehabilitation efforts, thereby contributing to a reduction in hospital stay durations. However, LLEs are controlled manually. This control method insufficiently engages the nervous system, therefore missing the opportunity of accelerating recovery further. Fortunately, recent advances in AI have made it possible to detect when a patient intends to move from their neurological signals to control an LLE. This paper compares three neurological signals served as inputs to such AI models for LLE control — electroencephalography (EEG), surface electromyography (sEMG), and a fusion of the two — based on accuracy, speed, and cost. sEMG outperforms EEG and EEG + sEMG in accuracy (96%), speed (59.9ms) and cost (US\$1,120), and is the best control input for wide clinical usage. EEG is the least performant in respect of all criteria due to its inherent noisiness. On the other hand, EEG + sEMG, having the average accuracy and speed of the two, offers optimal patient personalization in rehabilitation by leveraging both signals. However, this comparison is limited due to differences in the experiments analyzed. This development is significant because it can alleviate the financial burden for low- and middle-income patients undergoing rehabilitation.

Keywords — Neurorehabilitation, neurorobotics, lower limb exoskeleton, artificial intelligence, EEG, EMG.

I. INTRODUCTION

Neurorehabilitation is a medical process that helps patients with neurological conditions recover or increase functionality [1]. Patients with conditions such as spinal cord injury (SCI) or stroke often undergo neurorehabilitation to regain mobility and independence in daily activities. However, neurorehabilitation is costly both in time and money. The mean rehabilitation stay is 44 days, costing US\$26,340 out of a total of US\$70,601. Recently, lower limb exoskeletons (LLE) have been developed to tackle this issue by facilitating rehabilitation [2]. However, these robotic devices face a significant issue: they are controlled manually through buttons, which insufficiently engage the lower limb nervous system [2]. Activating the lower limbs can accelerate neuromuscular recovery, potentially helping patients walk again. Thus, this may shorten stays and reduce rehabilitation costs. Therefore, it is imperative to investigate alternative exoskeleton control inputs that simultaneously stimulate the nervous system and detect movement intention to accelerate recovery. By exploring alternatives, this rehabilitation may reach more people, especially those in low- and middle-income families.

The following research will discuss and compare different neurological signals for movement intention detection AI models used in exoskeleton control. These signals include electroencephalography (EEG), surface electromyography (sEMG), and a combination of the two (EEG + sEMG). Each input signal will be evaluated based on three key criteria: the accuracy of AI models trained on them, the speed of these models, and the cost of signal extraction equipment. This paper concludes that sEMG signals are a more efficient, faster, and more cost-effective method for detecting movement intention in LLE control compared to EEG or EEG + sEMG.

II. BACKGROUND

Despite recent advances in assistive lower limb exoskeletons, neurerehabilitation remains expensive and lengthy. The mean rehabilitation stay is 44 days, costing US\$16,129 for accommodation and US\$10,211 in opportunity cost¹. As a potential solution to reduce this period and its costs, LLEs assist movement in patients with lower movement disabilities [2]. An LLE is a wearable robotic device that helps movement for patients with lower motor disabilities [2, 3]. An example of an LLE can be seen in Fig. 1. LLE are especially helpful in rehabilitation, because they help physiotherapists move and train their patients without suffering from fatigue themselves. Currently, LLE movement is controlled through buttons or joysticks, which are proven to be robust for movement even for patients with complete paraplegia [2, 4]. While this control method is effective, it insufficiently engages the lower limb nervous system, thereby missing the opportunity to shorten rehabilitation [2]. As an alternative, recent research explores using neurological electrical signals generated by the body as input to AI models to classify whether a patient intends to move.

¹Opportunity cost is lost income potential. During rehabilitation, patients cannot work and thus face this cost.



Fig. 1. A physiotherapist conducting elastic band resistance training with her patient wearing an LLE [5].

A. Electroencephalography (EEG)

One such signal commonly used for LLE control is EEG. EEG is a technique that uses a cap with electrodes to record neuroelectric brain signals (see Fig. 2) [6]. These recorded signals can then be used by AI algorithms for movement intention detection [7-10]. However, processing these signals is time-consuming due to the need to remove artifacts. Artifacts are signals generated by physiological activities in various parts of the body — such as eye blinks, or heartbeats — that interfere with primary signals of interest (lower limb muscle and brain signals), making accurate analysis difficult [11]. To compensate for the extensive time required for EEG preprocessing, Gordleeva et al (2020) employ Linear Discriminant Analysis (LDA), a fast AI model well-suited for linear classification [7]. Its computational efficiency stems from the simplicity of linear operations, which involve straightforward calculations rather than complex transformations [12]. This makes LDA particularly advantageous for real-time or large-scale data processing. The principle of linear classification is illustrated in Fig. 4 (a).

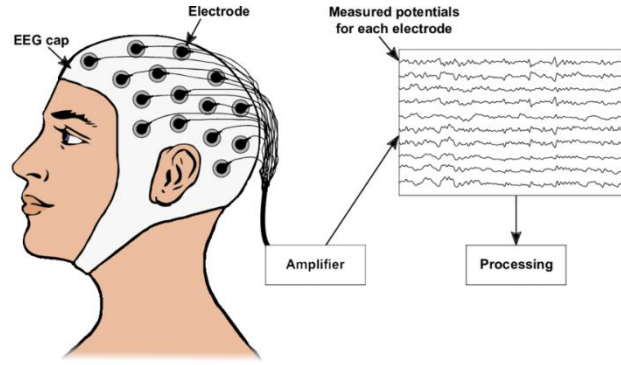


Fig. 2. EEG signal extraction and processing.

B. Surface Electromyography (sEMG)

Another neurological signal that serves as an LLE control input is sEMG. sEMG is a method of extracting musculoskeletal electric signals using electrodes placed on the surface of the skin [13]. Although sEMG relies on muscle movement, it can still be detected in semi-paraplegic patients, making it useful for neurorehabilitation [2]. However, the signals are often weaker in such patients, which may limit their effectiveness. Gaussian Kernel Linear Discriminant Analysis (GK-LDA), used by Qin et al. (2020), is an AI model that extends LDA to better handle the non-linearity of sEMG signals [8]. This model is a variation of LDA that uses a mathematical tool called a kernel, which allows the data to be viewed in a higher-dimensional space [14]. In this new space, patterns that are difficult or impossible to detect in the original form of the data become much easier to recognize. This makes GK-LDA especially useful for identifying the kinds of non-linear patterns (see Fig. 4) that are common in neurological signals.

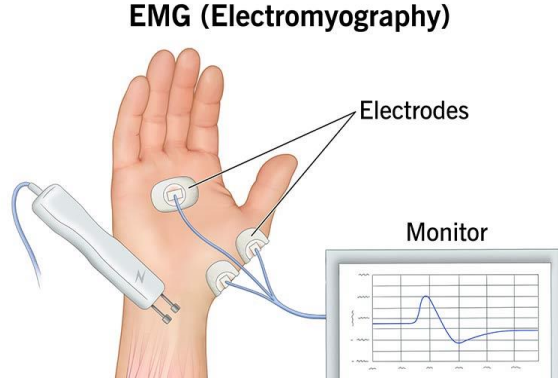


Fig. 3. EMG signal extraction through surface electrodes.

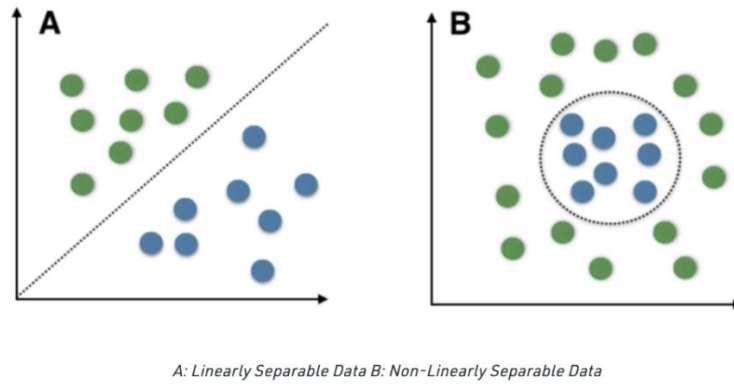


Fig. 4. Classification (separation) of linear (A) and non-linear (B) data. Generally, linear separation is easier than non-linear separation [15].

C. Fusion of EEG and sEMG

Recent research suggests that combining EEG and sEMG signals may offer more effective control of LLE than using either signal alone. This fusion, formed by bundling both signals before feeding them into a classifier, enhances model flexibility but slightly slows down intention detection [10]. Tortora et al. (2020) highlight its usefulness in adapting to different stages of rehabilitation, where EEG signals can compensate for weaker sEMG activity early on (see Fig. 5). Zhang et al. (2022) implement this approach using a combination of Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models, where CNNs filter out noise and LSTMs capture time-based signal changes [9]. While effective, these models require

significant computational resources and large datasets, making them slow to deploy for real time LLE control [16]. Nonetheless, this fusion approach shows strong potential for personalized and adaptable neurorehabilitation systems.

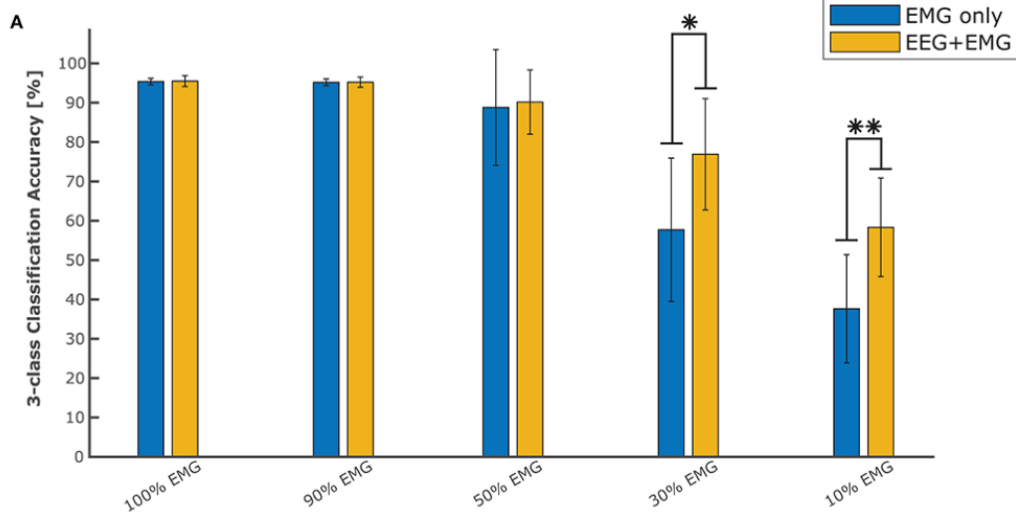


Fig. 5. Comparison of Accuracies of EMG, and EEG + EMG movement intention detection models. At weaker EMG levels (10-30%), EEG + EMG performs better [10].

D. Criteria for comparison

To determine which neurological signal is the most suitable input for movement intention detection AI models, accuracy, speed, and cost will be evaluated.

The first criterion, accuracy, is the measure of correct classifications made by an AI model. It is defined as the ratio of the correct number of classifications over the total number of classifications [17]:

$$Accuracy (\%) = \frac{\# \text{ Correct Classifications}}{\# \text{ Total Classifications}} \times 100\% \quad (1)$$

Classification is the task of separating data into different classes. In LLE movement intention detection, the goal is to correctly categorize signal data into 3 classes: right leg

movement, left leg movement, no movement. Usually, accuracy decreases as the number of classes grows because of the increase in non-linear combinations between classes [15].

The second criterion, speed, refers to how quickly an algorithm executes. This encompasses the time taken by all steps of a classification AI algorithm, as can be seen in Fig. 6.

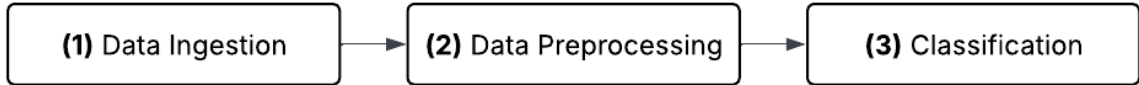


Fig. 6. Sensor data must be loaded **(1)**, then transformed into a better data structure **(2)**, and finally classified **(3)** [2, 17].

Speed is influenced by factors such as computational complexity and hardware, making direct comparisons between different experiments difficult. Although theoretical algorithm efficiency offers a fairer comparison, it is beyond this paper's scope. Thus, execution time in milliseconds (ms) will be evaluated while assuming comparable setups across experiments.

The third criterion, cost, corresponds to the average cost of signal extraction and processing equipment in US dollars. It includes the price of sensors, amplifiers and signal processing hardware. Conversely, it excludes the cost of computing equipment and specialized software. Because some data date back to 2020, this paper assumes that the cost of equipment has not changed over time. This paper focuses on wired systems because they are less expensive and suited for controlled rehabilitation environments where exoskeletons are used for training rather than daily use [18].

III. ANALYSIS

A. Accuracy

Model accuracy is essential for ensuring that the LLE correctly detects and responds to a user's movement intention.

Gordleeva et al. (2020) evaluated their EEG-based LDA model using data from 8 subjects, achieving 78.13% accuracy for binary classification (movement intention vs. no movement) and 51.13% accuracy for 3-class classification [7]. While these results exceed random chance (50% and 33% respectively), the performance remains limited due to the inherent noise and artifacts present in EEG signals, which pose fundamental challenges for motor imagery classification. In addition, due to the linearity of the LDA algorithm, noisy non-linear EEG signals do not fare well when the number of classes increases.

In contrast, Qin et al. (2020) reported substantially higher performance with their sEMG-based GK-LDA model, achieving 96% accuracy across 5 classes [8]. However, this comparison requires careful interpretation due to several methodological differences. First, Qin et al. (2020) used data from only 5 subjects compared to Gordleeva et al.'s (2020) 8 subjects, potentially introducing a selection bias where the model may have performed exceptionally well for this cohort but might not generalize to a broader population. Second, the subjects were all healthy in Qin et al.'s (2020) experiment. In practice however, patients suffering from neurological conditions have weak sEMG signals in the starting stages of neurorehabilitation.

Zhang et al.'s (2022) CNN-LSTM fusion model, trained on EEG and sEMG data from 8 subjects for 5-class classification, achieved 89.63% accuracy, approximately the average of individual EEG and sEMG performances [9]. However, the fusion approach substantially outperforms sEMG when sEMG signals are degraded. At 10% normal sEMG strength, the fusion model maintains 64.66% accuracy while the sEMG-only model drops to 39.33% [10]. Above 50% signal strength though, both models perform almost equivalently (see Fig. 6).

B. Speed

Speed is a key metric because the exoskeleton's movement must be natural and smooth for proper activation of the lower limb nerves. Gordleeva et. al's (2020) EEG model performed a classification every 500ms, though it is left unclear whether this frequency was set manually or if that is the natural speed of the algorithm [7]. Conversely, Qin et. al's (2020) sEMG model held a computation time of 59.9ms [8]. Finally, Zhang et. al's (2022) fusion model achieved an execution time of 265ms, effectively a midpoint between the EEG or EMG models alone [9]. The EEG and sEMG speeds are vastly different because EEG signals take longer to preprocess than sEMG signals. The delay is primarily due to the time-consuming process of EEG artifact removal, which also impacts the fusion model since it relies on EEG data.

C. Cost

To make it accessible, the cost of neurological signal extraction equipment should be reasonable. According to Ney et. al (2024), EEG collection equipment costs US\$9,181.15 and EEG cap electrodes cost US\$31.50 [19]. With an EEG cap having an average of 25 electrodes [20], the total electrode cost amounts to \$787.50. Thus, EEG equipment in total costs US\$9,968.65. Unfortunately, Ney et. al (2024) did not provide a detailed summary of the cost of different components in EEG signal extraction [19]. As such, it is difficult to assert whether this cost reflects the cost of an LLE EEG extraction setup.

On the other hand, sEMG sensors can cost as low as US\$112 thanks to recent research in open-source electronics [21]. In Qin et. al's study (2020), participants only had 3 sEMG sensors on each leg. As such, the total sensor cost would amount to US\$672. However, these open-source systems generally provide lower quality signals for significant analysis [21]. Sadly though, due to

companies' privacy on pricing though, it is hard to determine consumer price of sEMG systems through online investigation alone.

Finally, the EEG + sEMG hardware setup price would simply be the sum of the individual system prices. This would amount to US\$ 10,640.65. The hybrid setup allows users to tailor the system by prioritizing either EEG or sEMG components based on clinical needs or rehabilitation stage.

TABLE I
SUMMARY OF FINDINGS

Signal	Model	Number of Subjects	Number of Classes	Accuracy (%)	Speed (ms)	Cost (USD)
EEG	LDA	8	2	78.13	500	9,968.65
sEMG	GK-LDA	5	5	96.00	80	672
EEG + sEMG	CNN-LSTM	8	8	89.63	265	10,640.65

IV. CONCLUSION

This paper analyzed the accuracy, speed, and cost of three AI-based control inputs for LLE movement intention detection: EEG, sEMG, and EEG + sEMG. EEG-based models are inaccurate and slow due to noise generated by artifacts, while also requiring expensive acquisition equipment. Conversely, sEMG provides far superior accuracy and speed through cleaner signals, with reduced costs thanks to recent advances in open-source electrodes. In contrast, EEG + sEMG outperforms sEMG during early rehabilitation but performs comparably as rehabilitation progresses. However, this early-stage advantage comes with increased computational complexity and higher equipment costs.

That said, experimental variations in subject populations, classification tasks, model architecture, and equipment limit definitive cross-study comparisons. As such, a promising direction for future research is to conduct an experiment on real-life patients undergoing neurorehabilitation. Conducting the experiment under identical conditions — using the same equipment, signal preprocessing techniques, and AI models — would allow for clearer comparisons between EEG, sEMG, and EEG + sEMG. In addition, this study could offer more definitive evidence of sEMG's real-world performance, especially in paraplegic patients whose muscle recovery evolves over time.

In the end, sEMG offers an optimal balance of accuracy, speed, and cost-effectiveness, making it a strong candidate for LLE control in the later stages of rehabilitation. Leveraging sEMG as a control input may promote neural engagement, accelerate patient recovery, and reduce the high costs of clinical stays.

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